KubExporter

Export Kubernetes resources as clean, reproducible YAML/JSON

github.com/bakito/kubexporter



Run: `presenterm -x -X kubexporter/kubexporter.md`

Why KubExporter?

- **Backup / Disaster Recovery** – a plain-file snapshot of *every* resource in a cluster
- **Audit & Diff** – compare clusters or the same cluster over time
- **Migration** – move workloads between clusters / providers
- **Debugging** – grab the full state of a namespace for offline analysis
- **Manifest extraction** – turn hand-made cluster state into reusable manifests

Output is just files: keep them on disk, in a tar.gz, in S3 / GCS – or in Git, if that fits your workflow.

> kubectl get -o yaml gives you *noisy* output for *one* kind.
> KubExporter gives you *clean* output for *all* kinds – including CRDs.

Key Idea: exclude, don't include

Opt-out by default

- Discovers **all** API resources via the discovery API
- New CRDs are exported **automatically**
- You only say what you **don't** want

Clean output

- Strip runtime noise: `status`, `uid`, `resourceVersion`, `managedFields`, …
- Deterministic file names & sorted slices → **diff-friendly**



Getting Started

```
# Release binaries
https://github.com/bakito/kubexporter/releases

# Homebrew
brew tap bakito/tap && brew install --cask kubexporter

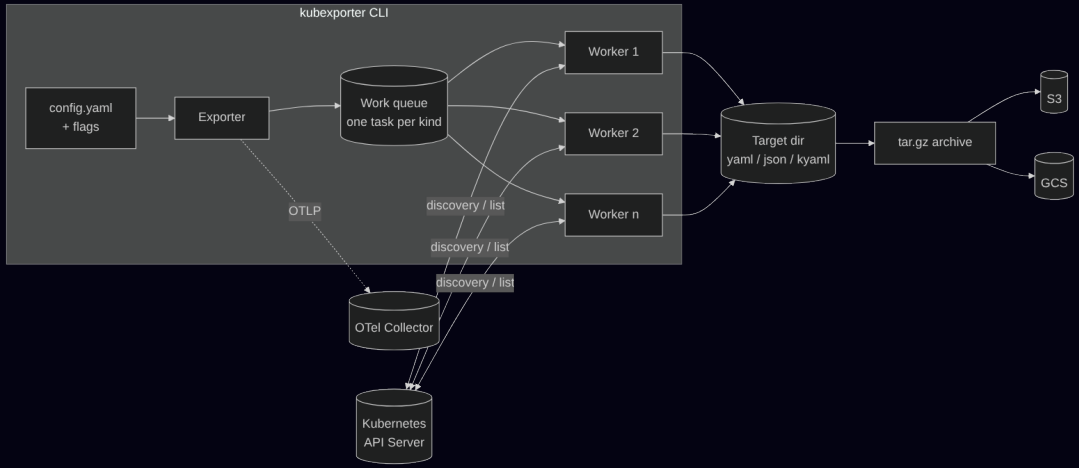
# kubectl plugin (krew)
kubectl krew install exporter
kubectl exporter --help
```

Uses your **kubeconfig** – all the usual flags work: `--context`, `--as`, `--token`, `--server`, …

```
kubexporter --version
kubexporter --help | sed -n '/Available Commands/,/^$/p'
```

snippet +exec is disabled, run with -x to enable

Architecture



- `-w / --worker` → parallel workers (one kind per task)
- `queryPageSize` → paged list requests for large clusters

Live Demo: first export

`demo-config.yaml` limits the export to `cert-manager` and `kube-system` (kinds without instances are filtered from the summary):

```
cd docs/presentation 2>/dev/null || true
kubexporter --config demo-config.yaml | grep -vE ' yes +0 +0 '
```

```
snippet +exec is disabled, run with -x to enable
```

Clean, diff-friendly manifests

Global & per-kind field excludes

```yaml
excluded:
  fields:
    - [status]
    - [metadata, uid]
    - [metadata, resourceVersion]
    - [metadata, managedFields]
  kindFields:
    Service:
      - [spec, clusterIP]
      - [spec, clusterIPs]
  preservedFields:
    fields:
      - [status, loadBalancer, ingress]
```

`preservedFields` → keep a child even if the parent is removed.

Result

```
bat
/tmp/kubexporter-demo/exports/kube-system/
Service.kube-dns.yaml
```

snippet +exec is disabled, run with -x to enable

Filtering resources

What	Config	Flag
Exclude kinds	`excluded.kinds`	`-e, --exclude-kinds`
Sane default excl.	–	`-d, --exclude-defaults`
Include only kinds	`included.kinds`	`-i, --include-kinds`
Namespaces	`namespace` / `namespaces`	`-n, --namespace`
+ cluster scoped	`includeClusterResources`	`--include-cluster-resources`
By field value	`excluded.kindByField`	–
By age	`createdWithin`	`--created-within 24h`
By owner	`considerOwnerReferences`	–

```yaml
excluded:
  kindByField:
    Secret:                          # drop Helm release secrets
      - field: [type]
        values: ['helm.sh/release.v1']
considerOwnerReferences: true        # skip resources whose owner is exported
```

Ad-hoc export with flags only

No config file needed — quick "what changed today?" snapshot:

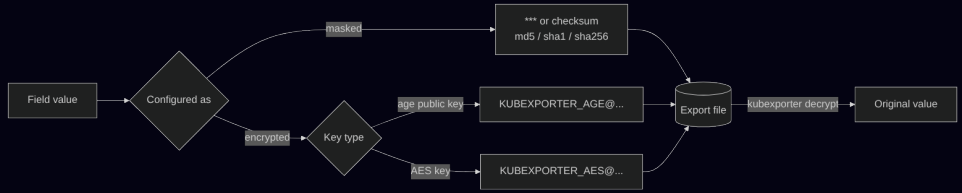
```
kubexporter -n kube-system -i ConfigMap,ServiceAccount \
  -t /tmp/kubexporter-demo/adhoc -c -p none --summary \
  --created-within 8760h
```

```
snippet +exec is disabled, run with -x to enable
```

```
ls -1 /tmp/kubexporter-demo/adhoc/kube-system | head -n 6
```

```
snippet +exec is disabled, run with -x to enable
```

Protecting sensitive data



- **Masking** – one-way: still lets you *detect changes* via checksum, never restore
- **Encryption** – reversible: exports can be stored or shared safely, restore when needed
- **age is preferred** over AES when both keys are configured

Encryption in the export

```
encrypted:
  aesKey: "12345678901234567890123456789012"   # or KUBEXPORTER_AES_KEY / K8s
Secret
  # agePublicKey: "age1 ... "                   # preferred: public-key crypto
  kindFields:
    Secret:
      - [data]
      - [stringData]
```

```
cut -c1-80
/tmp/kubexporter-demo/exports/cert-manager/Secret.cert-manager-webhook-ca.yaml |
bat -l yaml
```

snippet +exec is disabled, run with -x to enable

Decrypt (AES)

```
cp
/tmp/kubexporter-demo/exports/cert-manager/Secret.cert-manager-webhook-ca.yaml
/tmp/kubexporter-demo/secret.yaml
kubexporter decrypt --aes-key 12345678901234567890123456789012
/tmp/kubexporter-demo/secret.yaml
cut -c1-80 /tmp/kubexporter-demo/secret.yaml | bat -l yaml
```

snippet +exec is disabled, run with -x to enable

Keys can come from:

- flag `--aes-key` / env `KUBEXPORTER_AES_KEY` / interactive password prompt
- a Kubernetes Secret: `--aes-key-secret-namespace|name|key`

Encrypt & Decrypt with age

Encrypt existing files after the fact — only the recipient's **public** key is needed:

```
cd /tmp/kubexporter-demo
rm -f age.key && age-keygen -o age.key 2>/dev/null
PUB=$(age-keygen -y age.key)
kubexporter encrypt --age-public-key "$PUB" secret.yaml
cut -c1-80 secret.yaml | sed -n '2,4p' | bat -l yaml
```

snippet +exec is disabled, run with -x to enable

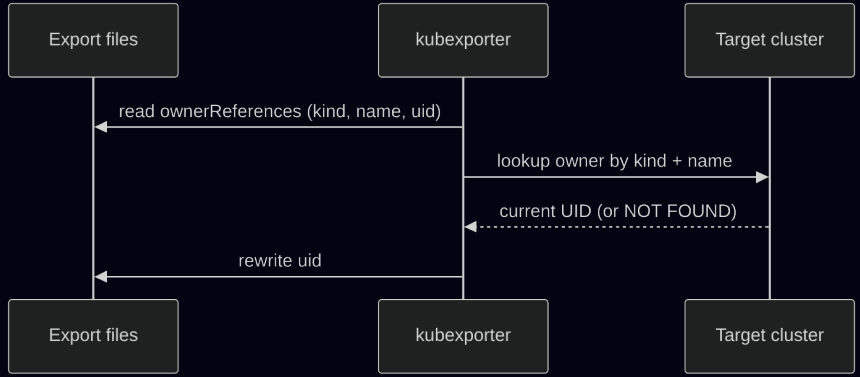
```
cd /tmp/kubexporter-demo
kubexporter decrypt --age-identity "$(grep AGE-SECRET-KEY age.key)" secret.yaml
```

snippet +exec is disabled, run with -x to enable

> Mixed files (AES **and** age values) are decrypted in a single pass.

Update Owner References

UIDs change when resources are re-created in another cluster → `ownerReferences` break.



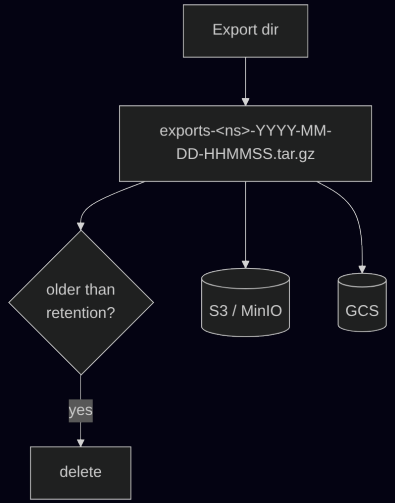

```
kubexporter update-owner-references
 FILE                                  OWNER KIND  OWNER NAME  UID FROM   UID
TO
 cert-manager/cilium.io.CiliumEndpoint…  Pod         cert-…      1d49…hhhh
```

Archives & Remote Storage

```yaml
archive: true            # -a
archiveTarget: /backups
archiveRetentionDays: 10 # rotate old
archives

s3:
  endpoint: minio.local:9000
  accessKeyID:  ...
  secretAccessKey:  ...
  secure: true
  bucket: k8s-backups

gcs:
  bucket: k8s-backups    # auth via ADC /
Workload Identity
```



Run it as a CronJob

```yaml
apiVersion: batch/v1
kind: CronJob
metadata:
  name: kubexporter
spec:
  schedule: "0 2 * * *"
  jobTemplate:
    spec:
      template:
        spec:
          serviceAccountName: kubexporter   # ClusterRole: get/list on *
          containers:
            - name: kubexporter
              image: ghcr.io/bakito/kubexporter:latest
              args: ["--config", "/config/config.yaml", "-p", "none", "--summary
"]
              envFrom: [{ secretRef: { name: kubexporter-keys } }]
          restartPolicy: OnFailure
```

- Image is published to ghcr.io and signed with **cosign**
- Encrypt with **age**: the Job only needs the *public* key
- Push archives to S3 / GCS and use archiveRetentionDays for rotation

Observability for SREs

```yaml
metrics:
  otlp:
    enabled: true      # --otlp-metrics
    endpoint: otel-collector:4317
    insecure: true
```

Auth headers via env:

```
KUBEXPORTER_METRICS_OTLP_HEADER_Authorization="Bearer <token>"
```

Grafana dashboard:
`examples/grafana/dashboard.json`

Metric
kubexporter.duration_seconds
kubexporter.errors
kubexporter.exported_resources
kubexporter.exported_size_bytes
kubexporter.namespaces
kubexporter.resource.instances
kubexporter.resource.query_duration_seconds
kubexporter.resource.export_duration_seconds

> Alert on errors > 0 or a missing run → your backup is silently broken.

UX: progress & summary

Mode	Use case
`bar`	default, interactive terminal
`bubbles`	fancy TUI (bubbletea)
`simple`	CI logs
`none`	CronJobs / scripting

- `--summary` → table per kind: instances, exported, query & export duration
- `--size` → size of the exported files
- `-v / --verbose` → list errors in summary, `-q / --quiet` → no output

```
kubexporter -n cert-manager -i ConfigMap,Secret -t /tmp/kubexporter-demo/ux -c \
  -p simple --summary --size 2>&1
```

snippet +exec is disabled, run with -x to enable

Live Demo: progress bar

The `bar` and `bubbles` modes need a real, interactive terminal.

`+acquire_terminal` gives the snippet the whole terminal. Press `ctrl+e` to run it, and press any key to go back to the slides.

```
clear
kubexporter --config demo-config.yaml -p bar --summary=false
echo
read -n 1 -s -r -p "Press any key to return to the presentation ..."
```

snippet +exec is disabled, run with -x to enable

```
clear
kubexporter --config demo-config.yaml -p bubbles --summary=false
echo
read -n 1 -s -r -p "Press any key to return to the presentation ..."
```

snippet +exec is disabled, run with -x to enable

Use case: diff two snapshots

```
cd /tmp/kubexporter-demo
kubexporter -n kube-system -i ConfigMap -t snap1 -c -q
kubectl -n kube-system create configmap kubexporter-demo --from-literal
=hello=world >/dev/null
kubexporter -n kube-system -i ConfigMap -t snap2 -c -q
diff -r snap1 snap2
kubectl -n kube-system delete configmap kubexporter-demo >/dev/null
```

snippet +exec is disabled, run with -x to enable

Clean fields + deterministic names + `sortSlices` = **only real changes show up**.

Recap

- Export **all** resources (incl. CRDs) – opt-out filtering
- Clean, deterministic, **diff-friendly** YAML / JSON / kyaml
- Fine-grained filters: kinds, namespaces, field values, age, owners
- **Mask** or **encrypt** (AES / age) sensitive fields – decrypt later
- Fix `ownerReferences` against a new cluster
- Archives with retention, upload to **S3 / GCS**
- **OTLP metrics** + Grafana dashboard for SREs
- Works as standalone binary, **kubectl plugin**, or **CronJob**

Try it: https://killercoda.com/bakito/scenario/kubernetes-kubexporter

Code: https://github.com/bakito/kubexporter